(*) On every Riemannian manifold, we will always use the Levi-Civita connection if there is no further comment.

1 Some Discussions in the Plane $\mathbb{R}^2$

1.1 Rotation Index

Let $\gamma : [a, b] \to \mathbb{R}^2$ be an admissible curve in the plane throught this subsection, i.e., γ is a piecewise smooth curve. We will call γ a **simple closed curve** if $\gamma(a) = \gamma(b)$ and $\gamma(t_1) \neq \gamma(t_2)$ for any $t_1 \neq t_2 \in [a, b)$. Denote

$$T(t) := \frac{\gamma'(t)}{|\gamma'(t)|}$$

hence we get a map $T : [a, b] \to \mathbb{S}^1$ (note that we have the canonical identification $T\mathbb{R}^2 \cong \mathbb{R}^2$).

If γ is a smooth curve, we define a **tangent angle function for** γ to be a continuous function

$$\theta : [a, b] \to \mathbb{R}$$

such that $T(t) = (\cos\theta(t), \sin\theta(t))$ for nay $t \in [a, b]$. To see why such funtion θ exists, just recall the theory about covering space (you can see [3] P146-P163 or [1] appendix A.54): consider the map $q : \mathbb{R} \to \mathbb{S}^1$, which is defined by $q(s) := (\cos s, \sin s)$, it's clear that q is a smooth covering map and the path-lifting property of covering maps ensure that there exists a continuous function $\theta : [a, b] \to \mathbb{R}$ such that $q \circ \theta = T$, i.e., we have the following commutative diagram

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "q"] \\ {[a,b]} \arrow[ur, "\theta"] \arrow[r, "T"'] & \mathbb{S}^1 \end{tikzcd}$$

hence by the unique lifting-property of covering space (you can see [1] appendix A.54(a)), such lift θ is uniquely determined by its single value on $[a, b]$, hence any two such lifts differs by a constant integer multiple of 2π, i.e., if θ_1 and θ_2 both satisfies the above diagram, then $\theta_1 - \theta_2 = 2k\pi$ with $k \in \mathbb{Z}$.

If γ is a simple closed differentiable curve with $\gamma'(a) = \gamma'(b)$, which means $(\cos\theta(a), \sin\theta(a)) = (\cos\theta(b), \sin\theta(b))$ and hence $\theta(a) - \theta(b) = 2k\pi$ $(k \in \mathbb{Z})$. Then we define the **rotation index of** γ to be

$$\rho(\gamma) := \frac{1}{2\pi}(\theta(b) - \theta(a))$$

it's clear that $\rho(\gamma) \in \mathbb{Z}$, and by our discussion above, we know that ρ is well-defined, i.e., independent of the choice of the lift θ.

1.2 Jumps at Vertices

Now, let $\gamma : [a, b] \to \mathbb{R}^2$ be an admissible simple closed curve, let $(a_0, a_1, \ldots, a_k)$ be an admissible partition of $[a, b]$, i.e., $\gamma|_{[a_{i-1}, a_i]}$ is smooth. These points $\gamma(a_i)$ are called the vertices of γ, and the curve segments $\gamma|_{[a_{i-1}, a_i]}$ are called its **edges**. At each vertex $\gamma(a_i)$, γ has left-hand and right-hand velocity vectors denoted by $\gamma'(a_i^-)$ and $\gamma'(a_i^+)$ respectively. At each vertex, we define the **exterior angle at** $\gamma(a_i)$ to be the angle ε_i, which is the angle from $T(a_i^-)$ to $T(a_i^+)$, chosen to be $\varepsilon_i \in (-\pi, \pi)$, with a positive sign if $(T(a_i^-), T(a_i^+))$ is an standard oriented basis of $\mathbb{R}^2$ and with negative sign otherwise, where $T(a_i^+) = \frac{\gamma'(a_i^+)}{|\gamma'(a_i^+)|}$ and $T(a_i^-) = \frac{\gamma'(a_i^-)}{|\gamma'(a_i^-)|}$ just as what we have defined. The **inetrior angle at** $\gamma(a_i)$ is defined as $\theta_i := \pi - \varepsilon_i$. (see the following figure)

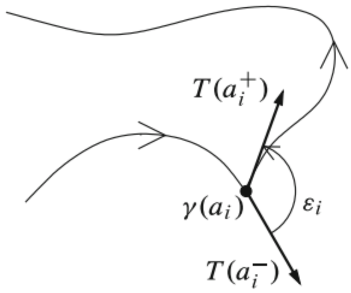

图 1: an interior angle

1.3 Curved Polygons in the Plane

For an admissible simple closd curve $\gamma : [a, b] \to \mathbb{R}^2$, we call a vertex $\gamma(a_i)$ a **cusp vertex** if $T(a_i^-) = -T(a_i^+)$. (see the figure below)

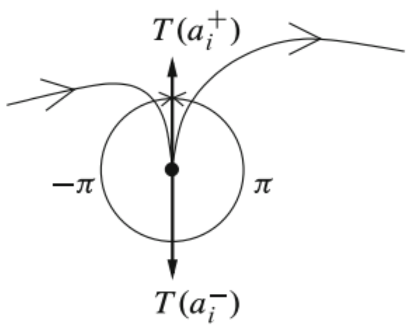

图 2: a cusp vertex

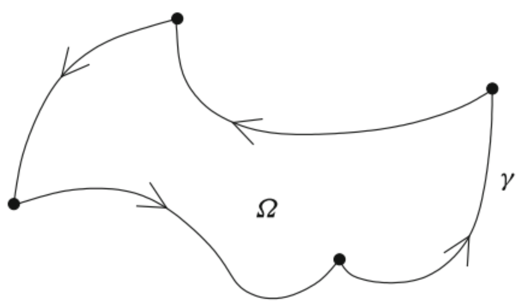

图 3: a positively oriented curved polygon

Curved Polygon in the Plane

Definition 1.1. : A **curved polygon** is an admissible simple closed curve in the plane without cusp vertices, whose image is the boundary of a precompact open set $\Omega \subseteq \mathbb{R}^2$.

Remark: If γ' is ppsitively oriented with repect to the induced orientation on $\partial\Omega$ in the sense of Stokes's theorem, we say that γ is positively oriented.

1.4 Rotaion Index for a Curved Polygon

Now, we define the **tangent angle function for a curved polygon** γ to be a piecewise continuous function $\theta : [a, b] \to \mathbb{R}$ such that $T(t) = (\cos\theta(t), \sin\theta(t))$ at each point where γ is smooth. More over, we have

$$\theta(a_i) = \lim_{t \to a_i^-} \theta(t) + \varepsilon_i, \quad \theta(b) = \lim_{t \to b^-} \theta(t) + \varepsilon_k$$

Hence, we can define the **rotation index for a curved polygon as**

$$\rho(\gamma) := \frac{1}{2\pi}(\theta(b) - \theta(a))$$

just as in the smooth case.

1.5 Rotation Index Theorem for Curved Polygons in Plane

Rotation Index Theorem (Umlaufsatz)

Theorem 1.1. : *The rotation index of a positively oriented curved polygon in the plane is* $+1$.

Proof: For the details of the proof, you can see [1] P267-P271. □

2 Gauss-Bonnet Theorem

We restrict our attention to the case of an oriented Riemann 2-maniold (M, g).

2.1 Curved Polygons in M

An admissible simple closed curve $\gamma : [a, b] \to M$ is called a **curved polygon in** M if the image of γ is a boundary of a precompact open set $\Omega \subseteq M$ and there exists an oriented smooth coordinate disk containing $\overline{\Omega}$ under whose image γ is a curved polygon in $\mathbb{R}^n$. (see the figure below)

In particular, if all the edges of such polygon are geodesics, we call it a **geodesic polygon**.

For a curved polygon in M, the definitions discussioned above go unchanged on it.

2.2 Rotation Index Theorem for Curved Polygons in M

Suppose $\gamma : [a, b] \to M$ is a curved polygon and $\gamma = \partial\Omega$, let (U, φ) be an oriented smooth chart such that $\overline{\Omega} \subseteq U$. We use this coordinate map φ to transfer γ, Ω and g to the plane, hence we

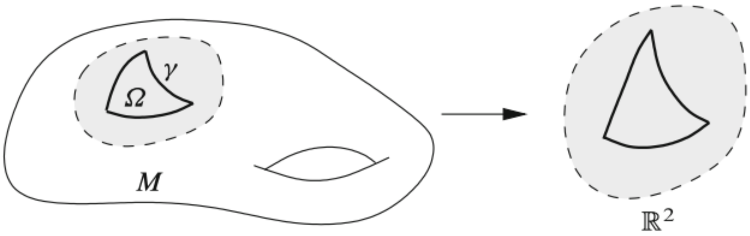

图 4: a curved polygon on a surface

may assume g is a metric on some open subset $\widehat{U} \subseteq \mathbb{R}^2$ and γ is a curved polygon in $\widehat{U}$. Apply the Gram-Schmdit algorithm to (∂_x, ∂_y), we get the oriented orthonormal frame (E_1, E_2) for g on $\widehat{U}$.

The **tangent angle function for** γ is the piecewise continuous function $\theta : [a, b] \to \mathbb{R}$ such that

$$T(t) = \cos\theta(t) E_1|_{\gamma(t)} + \sin\theta(t) E_2|_{\gamma(t)}$$

at each $t \in [a, b]$ where γ' is continuous. Just as in the planar case.

The **rotation index for** γ is

$$\rho(\gamma) = \frac{1}{2\pi}(\theta(b) - \theta(a))$$

Rotation Index Theorem for Curved Polygons in M

Lemma 2.1. *: If (M, g) is an oriented Riemannian manifold of dimension 2, the rotation index of any curved polygon is +1.*

Proof (sketch): Let $\overline{g}$ denote the Euclidean metric in $\mathbb{R}^2$. For $\forall s \in [0, 1]$, define the metric

$$g_s := sg + (1 - s)\overline{g}$$

if we use the given coordinate oriented chart to regard γ as a curved polygon in $\mathbb{R}^2$, then we can compute its angle function either w.r.t g or w.r.t. $\overline{g}$. In both cases, since $\theta(b), \theta(a)$ represent the angle between the same two vectors, we have $\rho_g(\gamma) \in \mathbb{Z}$ and $\rho_{\overline{g}}(\gamma) \in \mathbb{Z}$. Thus for the same reason, we can know that

$$\rho_{g_s}(\gamma) \in \mathbb{Z}$$

for $\forall s \in [0, 1]$, which means the function $f(s) := \rho_{g_s}(\gamma) \in \mathbb{Z}$ for all $s \in [0, 1]$. You can check that $f(s)$ is actually a continuous function on $[0, 1]$. Because f is a continuous and integer-valued function, hence we have $\rho_g(\gamma) = f(1) = f(0) = \rho_{\overline{g}}(\gamma) = 1$, which is obtained from theorem 1.1 □

Remark: By this lemma, we see that the rotation index of a curved polygon in M is independent of the choice of the coordinates.

2.3 Gauss-Bonnet Formula

For a curved polygon γ in M, we assume for convenience that $\gamma' \equiv 1$ since every curve has unit-speed reparametrization, in such case $T(t) = \gamma'(t)$. More over, there is a unique unit normal vector field N along the smooth portions of γ such that $(\gamma'(t), N(t))$ is a positively oriented orthonormal basis of $T_{\gamma(t)}M$. If γ is positively oriented as the boundary of Ω, this is equivalent to N being the inward-pointing normal to $\partial\Omega$.

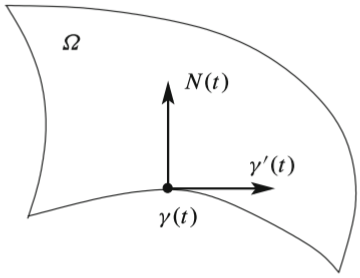

图 5: $N(t)$ is the inward-pointing normal

Singed Curvature

Definition 2.1. : Let γ and N be as in the discussion above, the **singed curvature of** γ is defined to be

$$\kappa_N(t) := \langle D_t\gamma'(t), N(t)\rangle_g$$

Remark: Since $\langle\gamma'(t), \gamma'(t)\rangle \equiv 1$, we have

$$\frac{d}{dt}\langle\gamma'(t), \gamma'(t)\rangle = 2\langle D_t\gamma'(t), \gamma'(t)\rangle \equiv 0$$

which means that $D_t\gamma'(t)\perp\gamma'(t)$ for any t, and therefore we get $D_t\gamma'(t) = \kappa_N(t)N(t)$. Hence in this case, the geodesic curvature of γ is $\kappa(t) := |D_t\gamma'(t)| = |\kappa_N(t)|$.

Gauss Bonnet Formula

Theorem 2.1. *: Let (M, g) be an oriented Riemannian 2-manifold, let γ be a positively oriented polygon in M and $\partial\Omega = \gamma$, let $\gamma(a_i)$ $(i = 1, \ldots, k)$ be its vertices, then we have*

$$\int_\Omega K dA + \int_\gamma \kappa_N ds + \sum_{i=1}^{k} \varepsilon_i = 2\pi$$

where K is the Gaussian curvature of g, dA is the Riemannian volume form, $\varepsilon_i(i = 1, 2, \ldots, k)$ are the exterior angles of γ, the integral $\int_\gamma$ is taken with respect to the arc length.

Proof: Let (x, y) be oriented smooth local coordinate on $U \supseteq \overline{\Omega}$, let $\theta : [a, b] \to \mathbb{R}$ be a tangent angle function of γ, by Lemma 2.1, we get

$$2\pi = \theta(b) - \theta(a) = \sum_{i=1}^{k} \varepsilon_i + \sum_{i=1}^{k} \int_{a_{i-1}}^{a_i} \theta'(t)dt$$

Apply Gram-Schmdit algorithm to ∂_x and ∂_y, we get the positively oriented g-orthonormal frame (E_1, E_2). Then we have

$$\gamma'(t) = \cos\theta(t)E_1|_{\gamma(t)} + \sin\theta(t)E_2|_{\gamma(t)}$$

$$N(t) = -\sin\theta(t)E_1|\gamma(t) + \cos\theta(t)E_2|_{\gamma(t)}$$

thus we can get

$$D_t\gamma' = -(\sin\theta)\theta' E_1 + (\cos\theta)\nabla_{\gamma'}E_1 + (\cos\theta)\theta' E_2 + (\sin\theta)\nabla_{\gamma'}E_2$$

$$= \theta' N + (\cos\theta)\nabla_{\gamma'}E_1 + (\sin\theta)\nabla_{\gamma'}E_2$$

Note that $\langle E_i, E_j\rangle = \delta_{ij}$ $(i,j = 1,2)$, so for any $v \in T_{\gamma(t)}M$, we have

$$\nabla_v|E_1|^2 = 2\langle \nabla_v E_1, E_1\rangle = 0$$

$$\nabla_v|E_2|^2 = 2\langle \nabla_v E_2, E_2\rangle = 0$$

$$\nabla_v\langle E_1, E_2\rangle = \langle \nabla_v E_1, E_2\rangle + \langle E_1, \nabla_v E_2\rangle = 0$$

these euqalities implies that

$$\nabla_v E_1 \,//\, E_2, \quad \nabla_v E_2 \,//\, E_1$$

now we define

$$\omega(v) := \langle E_1, \nabla_v E_2\rangle = \langle \nabla_v E_1, E_2\rangle \text{ for } \forall v \in T_{\gamma(t)}M$$

it's easy to see that ω is $C^\infty(M)$-linear over v, which means that ω is a 1-form. We get

$$\nabla_v E_1 = -\omega(v)E_2$$

$$\nabla_v E_2 = \omega(v)E_1$$

which is also saying

$$\nabla_v \begin{pmatrix} E_1 & E_2 \end{pmatrix} = \begin{pmatrix} E_1 & E_2 \end{pmatrix} \begin{pmatrix} 0 & \omega(v) \\ -\omega(v) & 0 \end{pmatrix}$$

By the definition of κ_N, we obtain

$$\kappa_N = \langle D_t\gamma', N\rangle = \langle \theta' N, N\rangle + \cos\theta\langle \nabla_{\gamma'}E_1, N\rangle + \sin\theta\langle \nabla_{\gamma'}E_2, N\rangle$$

$$= \theta' - \cos\theta\langle \omega(\gamma')E_2, N\rangle + \sin\theta\langle \omega(\gamma')E_1, N\rangle$$

hence we have

$$2\pi = \theta(b) - \theta(a) = \sum_{i=1}^{k}\varepsilon_i + \sum_{i=1}^{k}\int_{a_{i-1}}^{a_i}\theta'(t)dt = \sum_{i=1}^{k}\varepsilon_i + \sum_{i=1}^{k}\kappa_N(t)dt + \sum_{i=1}^{k}\int_{a_{i-1}}^{a_i}\omega(\gamma'(t))dt$$

note that $|\gamma'(t)| = 1$, so we get

$$2\pi = \sum_{i=1}^{k}\varepsilon_i + \int_\gamma \kappa_N ds + \int_\gamma \omega$$

(for more details in line integrals on manifolds, you can see [4] P287-P291)

Since $\partial\Omega = \gamma$, by Stokes' theorem, we get

$$\int_\gamma \omega = \int_\Omega d\omega$$

we now show that $d\omega = KdA$:

Note that (E_1, E_2) are orthonormal frame, by the property of Riemannian volume form, we get $dA(E_1, E_2) = 1$ (for this peoperty, you can see [4] P389 or [1] P30). Hence

$$KdA(E_1,E_2) = K = Rm(E_1,E_2,E_2,E_1) = \langle \nabla_{E_1}\nabla_{E_2}E_2 - \nabla_{E_2}\nabla_{E_1}E_2 - \nabla_{[E_1,E_2]}E_2, E_1\rangle$$

$$= \langle \nabla_{E_1}(\omega(E_2)E_1) - \nabla_{E_2}(\omega(E_1)E_1) - \omega([E_1,E_2])E_1, E_1\rangle$$

$$= \langle E_1(\omega(E_2))E_1 + \omega(E_2)\nabla_{E_1}E_1 - E_2(\omega(E_1))E_1 - \omega(E_1)\nabla_{E_2}E_1 - \omega([E_1,E_2])E_1, E_1\rangle$$

note that $\nabla_{E_1}E_1 \,//\, E_2$ and $\nabla_{E_2}E_1 \,//\, E_2$, so we have

$$KdA(E_1,E_2) = E_1(\omega(E_2)) - E_2(\omega(E_1)) - \omega([E_1,E_2]) = d\omega(E_1,E_2)$$

so it leads to $KdA = d\omega$ since they are both bilinear. This gives us

$$2\pi = \sum_{i=1}^{k} \varepsilon_i + \int_\gamma \kappa_N ds + \int_\Omega KdA$$

which is the desired equation. □

Angle-Sum Theorem

Corollary 2.1. : *The sum of the interior angles of a Euclidean triangle is π.*

Proof: Just need to note that $K \equiv 0$, $\kappa_N \equiv 0$ and $k = 3$ in this case. □

Remark: So we know that Gauss-Bonnet formula is the generalization of the usual angle-sum theorem in plane geometry.

2.4 Gauss-Bonnet Theorem

Let M be a smooth compact 2-manifold, a **curved triangle in** M is a curved polygon in M with 3 edges and 3 vertices.

Triangulation

Definition 2.2. : A **smooth triangulation of** M is a finite collection of curved triangles with disjoint interiors such that the union of the triangles with their interior is M, and the intersection of any pair of triangles (if nonempty) is either a simple vertex of each or a single edge of each.

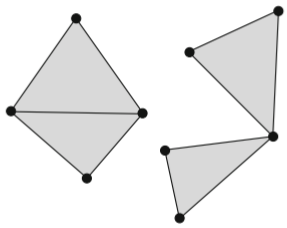

图 6: valid inersections

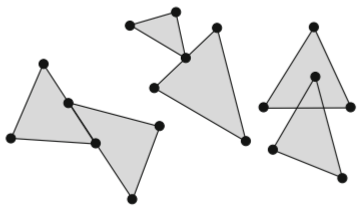

图 7: invalid inersections

By the theory of topology, every smooth, compact surfaces admits a smooth triangulation, even more general, every compact topological 2-manifold admits a triangulation without the assumption of smoothness, which is porved by Tibor Rado in 1925.

Euler Characteristic

Definition 2.3. : If M is a triangulated 2-manifold, the **Euler charcteristic of** M (with respect to the given triangulation) is defined to be

$$\chi(M) := V - E + F$$

where V is the number of vertices in the triangulation, E is the number of edges and F is the number of faces (the interior of the triangles).

Remark: It is well-known in the theory of algebraic topology that the Euler characterstic is actually a topological invariant, which means $\chi(M)$ is independent of the choice of the triangulation (for this you can see [5] P146-P147 or [3] P193-P194).

Now, we going to give the most important result in this note.

Gauss-Bonnet Theorem

Theorem 2.2. *: If (M, g) is an orientable compact Riemannian 2-manifold which is smoothly triangulated, then we have*

$$\int_M K dA = 2\pi\chi(M)$$

where K is the Gaussian curvature of g and dA is the Riemannian volume form.

Proof: We just need to prove the case that M is connected, because if it is not, we can prove this theorem for each connected component and add up the results.

Let $\{\Omega_i | \ i = 1, 2, \dots, F\}$ denote the faces of the triangulation. For each i, let $\{\gamma_{ij} | \ j = 1, 2, 3\}$ denote the edges of Ω_i, let $\{\theta_{ij} | \ j = 1, 2, 3\}$ denote its interior angles, hence $\theta_{ij} = \pi - \varepsilon_{ij}$ for any $i = 1, 2, \dots, F$ and $j = 1, 2, 3$. Apply the Gauss-Bonnet formula to each triangle and add them up, we get

$$\sum_{i=1}^{F} \int_{\Omega_i} K dA + \sum_{i=1}^{F} \sum_{j=1}^{3} \int_{\gamma_{ij}} \kappa_N ds + \sum_{i=1}^{F} \sum_{j=1}^{3} (\pi - \theta_{ij}) = \sum_{i=1}^{F} 2\pi$$

here note that every edge integral actually appears twice with opposite direction, which implies that

$\sum_{i=1}^{F}\sum_{j=1}^{3}\int_{\gamma_{ij}}\kappa_N ds = 0$, so

$$\sum_{i=1}^{F}\int_{\Omega_i} KdA + \sum_{i=1}^{F}\sum_{j=1}^{3}(\pi - \theta_{ij}) = \int_M KdA + 3\pi F - \sum_{i=1}^{F}\sum_{j=1}^{3}\theta_{ij} = 2\pi F$$

here, note that every inerior angle θ_{ij} appears once and the angles that touch the common vertex add up to 2π (see the figure below).

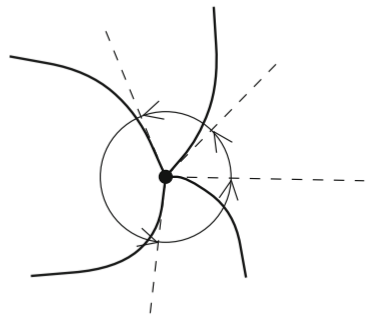

图 8: interior angles at a vertex add up to 2π

from this, we can get $\sum_{i=1}^{F}\sum_{j=1}^{3}\theta_{ij} = 2\pi V$, therefore, we get

$$\int_M KdA = 2\pi V - \pi F$$

Finally, we just need to observe that each egde appears precisely in two triangles and each triangle has exactly three egdes, this gives us the formula

$$2E = 3F$$

so we get $F = 2E - 2F$, hence we obtain

$$\int_M KdA = 2\pi V - \pi(2E - 2F) = 2\pi\chi(M)$$

this is what we want. □

Remark 1: In fact, if M is not orientable, this theorem can also be proved, but we need to use the concept about **Riemannian densities**. (for the details you can see [4] P432-P434)

Reamrk 2: By the classification theorem in topology, we know that any compact, connected, oritentable Riemannian 2-manifold M is homeomorphic to a sphere or a connected sum of n tori; any non-orientable one is homeomorphic to a connected sum of n copies of the real projective plean $\mathbb{RP}^2$ (for the proof about this theorem, you can see [3] P92-P101 or [6] P174& P267). The number n is called the **geneus** of M.

Remark 3: Gauss-Bonnet theorem is a local-to-global theorem in Riemannian geometry, the Gaussian curvature clearly depends on the differenial structure on M, but amazingly, by doing the interal on the left-hand side, we get a topological invariant on the right-hand side, which is independent of the differential structure on M!

2.5 Gauss-Bonnet-Chern Theorem

There is a generalization of Gauss-Bonnet theorem that deserves a mention. For a compact manifold M of any dimension, the Euler characteristic $\chi(M)$ can be defined analogously to that of a surface and is a topological invariant. (see [6] P373). It turns out that it is always zero for an odd-dimensional compact manifold, but it is a nontrivical invariant in each even-dimensional case.

The **Gauss-Bonnet-Chern theorem** equates the Euler characteristic of an even dimensional compact oriented Riemannian manifold to a certain curvature integral. In 1925, Heinz Hopf proved this theorem in the version of an embedded Riemannian huypsersurface in Euclidean space, and independently proved by Carl Allendoerfer and Warner Fenchel in 1940 for an embedded Riemannian submanifold of any Euclidean space (this was befroe Nash's 1956 proof that every Riemannian manifold has such an embedding). Finnally, an intrinsic proof for the general case was discoverd by Shiing-Shen Chern (陈省声) in 1944 for a complete discussion.

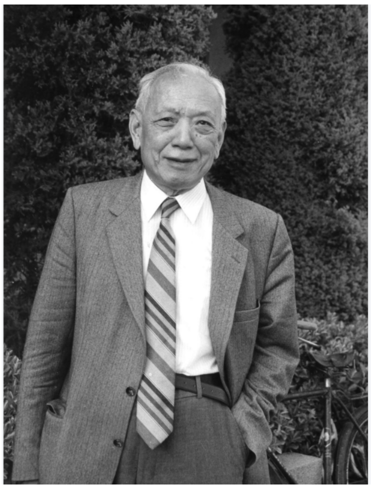

图 9: 陈省身

The theorem uses the theoy of characteristic class, which is written as

$$\int_M \mathrm{Pf}(Rm) = (2\pi)^n \chi(M)$$

where M is any $2n$-dimensional oriented compact Riemannian manifold, Pf is called the **Pfaffian**. (you can refer to [7] P230-P233)

参考文献

[1] John M.Lee *Introduction to Riemannain manifolds*

[2] 陈维桓, 李兴校 黎曼几何引论 *(上册)*

[3] 尤承业 基础拓扑学讲义

[4] John M.Lee *Introduction to Smooth manifolds*

[5] Allen Hatcher *Algebraic Topology*

[6] John M.Lee *Introduction to Topological manifolds*

[7] Loring W.Tu *Differential Geomtry*